INCH-POUND

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PERFORMANCE SPECIFICATION

ELECTRONIC TEST EQUIPMENT, INTERMITTENT FAULT DIAGNOSTIC (ELECTRICAL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the minimum performance requirements for equipment to detect and isolate nanosecond, microsecond, and millisecond intermittent faults (see 6.5.10), which can occur in conductive paths (see 6.5.2). Conductive paths include Line Replaceable Unit (LRU)/Weapon Replaceable Assembly (WRA) (see 6.5.12 and 6.5.19) chassis and backplane circuits and their wire harnesses; weapon system Electrical Wiring Interconnect System (EWIS) (see 6.5.4 and 6.5.20); and patch cables, electronic test cables and their connectors. This specification is not intended to address hard opens (see 6.5.14), shorts (see 6.5.16), nor constant function failures found in routine electronics repair.

1.2 Classification. All intermittent fault diagnostic equipment must be able to detect and isolate intermittent faults on any and all Unit Under Test (UUT) conductive paths. Diagnostic equipment is classified by its intermittent fault duration detection and isolation (see 3.10.1.1) capability in LRU/WRA backplane circuits and associated wire harnesses and in weapon systems' EWIS, as follows:

Category 1. Short duration intermittent faults (see 6.5.17) that are under 100 nanoseconds and typically have resistance variations of 10 ohms or less.

Category 2. Intermediate duration intermittent faults (see 6.5.9) that are 101 nanoseconds to 500 microseconds and typically have resistance variations between 10 and 500 ohms.

Category 3. Long duration intermittent faults (see 6.5.13) that are 501 microseconds to 5 milliseconds and typically have variations of 501 ohms or more.

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to LKE-NAVAIRDocComments@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-28800	-	Test Equipment for Use with Electrical and Electronic Equipment

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification marking of U.S. Military Property
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-1472	-	Human Engineering

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-527	-	Guidance for Intermittent Fault Emulator (IFE)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS

29 CFR 1910.1200	-	Occupational Safety and Health Standards
40 CFR 82	-	Protection of Stratospheric Waste
40 CFR 261	-	Identification and Listing of Hazardous Waste
40 CFR 355	-	Emergency Planning and Notification
40 CFR 372.65	-	Specific Toxic Chemical Listings
49 CFR 173	-	General Requirements for Shipments and Packaging

(Copies of these documents are available online at https://www.govinfo.gov/app/collection/cfr/.)

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DEPARTMENT OF DEFENSE INSTRUCTIONS

DoDI 8500.01	-	Cybersecurity
DoDI 8510.01	-	Risk Management Framework (RMF) for DoD Information Technology (IT)

(Copies of these documents are available online at https://dodcio.defense.gov/Library/.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern. Individual item, as used in this specification, refers to the intermittent fault diagnostic equipment.

3.2 First article. When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with 4.3. The number of samples should be as defined by the procuring activity (see 6.2g).

3.3 Diagnostic equipment capability. Equipment capability shall be as specified in the applicable specification sheet. The specification sheet shall specify the intermittent fault duration detection category (see 1.2) and additional capabilities required to detect and isolate intermittent faults (for applicability of diagnostic equipment, see 6.4).

3.4 Application. The diagnostic equipment shall interface with the input/output connections such as connectors and terminal boards of a replaceable package of avionic equipment LRU/WRA or weapon system EWIS. For the purposes of this specification, EWIS only refers to electrical interconnections and not fiber optics. The function of the diagnostic equipment is to troubleshoot the LRU/WRA or weapon system EWIS by detecting and isolating (see 6.5.8) intermittent faults. The LRU/WRA chassis conductive paths include all of the electrical components that transmit the signal or power from the LRU/WRA input/output connections to the input/output connections of replaceable packages (commonly referred to as the SRA/SRU (shop replaceable assemblies/shop replaceable units [see 6.5.15]) in the LRU/WRA. For chassis/backplane diagnostic troubleshooting, the SRA is typically removed from the LRU/WRA. For the SRA and/or entire LRU/WRA troubleshooting, the designated automatic test equipment (ATE) should be used. The weapon system EWIS includes all of the electrical components that transmit the signal or power, such as connectors, contacts, metallic wire, braid, terminals, splices, contactors, relays, circuit breakers, switches, and other items, materials, and processes, designed, integrated, and fabricated to provide the weapon system electrical and electronic interconnecting network. For weapon system EWIS troubleshooting beyond the detection and isolation of intermittent faults, designated test equipment should be used. The susceptibility of LRU/WRA or weapon system EWIS to functional failures caused by intermittent faults should be considered when detecting and isolating intermittent faults (see 6.5.11).

3.5 Detail requirements. The diagnostic equipment shall comply with all the requirements specified herein and the applicable specification sheet.

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3.6 Material. The material used shall enable the diagnostic equipment to meet the performance requirements of this specification and the applicable specification sheet.

3.7 Operating environment. The operating environment of the diagnostic equipment shall be specified in the applicable specification sheet.

3.8 Design and construction. Diagnostic equipment shall be constructed with parts and materials designed to provide the performance, reliability, and service life under the environmental and operating conditions specified herein. Static discharge control shall be provided for the protection of electronic devices during assembly and handling. Personnel safety and connector reliability shall be provided by ensuring that exposed interface connectors use scoop-proof design and receptacles use socket contacts. Captive metal connector covers shall be provided for all Support Equipment (SE) employed in non-environmentally protected environments (e.g., flight deck or aircraft hangar).

3.8.1 User interface. Diagnostic equipment shall be designed using an open-systems architecture approach, including the use of commercially available non-proprietary software. An open system is one that uses well-established, non-proprietary standards for interfaces, services, hardware, software, and supporting formats (e.g., Microsoft Excel, LabVIEW, USB, etc.). This enables components to be utilized across a wide range of systems, to interoperate with other components on local and remote systems, and to interact with users in a manner that facilitates portability. If third-party/proprietary software is employed, it should be built using mainstream applications such as Visual Studio, etc., and used, where required, for the interchange of data/communications using common formats such as pdf, csv, xls file formats, and USB, Ethernet, Bluetooth, etc.

3.8.2 Expandability. The diagnostic equipment shall include expansion capability (see 6.2.u). The components' expandability shall be as specified in the applicable detail specification sheet.

3.9 System physical characteristics (see 6.9).

3.9.1 Size and weight. The individual diagnostic equipment components' weight and size shall be as specified in the applicable specification sheet. The final size and weight shall be approved by the cognizant procuring activity.

3.9.2 Interface cables. The diagnostic equipment shall be capable of interfacing with LRU/WRA or weapon system EWIS through the use of interface cabling to connect the diagnostic equipment input/output connectors to the LRU/WRA or weapon system EWIS input/output connectors. Interface cabling may directly connect the diagnostic equipment to the LRU/WRA or weapon system EWIS, use existing interface cabling, or use an adapter cable to interface with existing interface cabling. Existing interface cabling is often referred to as Interface Test Adapters (ITAs), Interface Adapter Cable Sets (IACS), or Test Adapter Cables (TACs). Applicable interface cables shall be identified by the purchaser in the acquisition requirements (see 6.2v). Diagnostic equipment interface connectors shall continue to function when subjected to the environmental conditions specified herein and in the applicable specification sheet. All exposed connectors shall be fitted with captive metal caps or plugs (see 3.8).

3.9.2.1 Existing Test Program Set (TPS) usage. If the use of existing TPS is required, it shall be specified by the procuring activity (see 6.2.v). The TPS typically consists of an adapter cable, software, and documentation. See 6.6 for standard connector interface configuration.

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3.9.3 Input power (see 4.7.1). The diagnostic equipment shall operate at 50, 60 Hz single phase with voltage selectable from 100/120/220/240 volts Root Mean Square (RMS), unless otherwise specified in the applicable specification sheet.

3.9.3.1 Interruption of power source (see 4.7.2 and 4.7.3). Voltage or frequency interruptions shall not cause damage to the equipment. A power source interruption is when the voltage falls in the range of 0 to 70 percent of nominal or frequency falls in the range of 0 to 90 percent of nominal. Equipment performance is not required during the interruption of the power source. After power interruptions of any duration, the equipment shall automatically resume operation at either the last operating condition before the interruption or a default power-up condition.

3.10 Performance characteristics. Diagnostic equipment shall perform electrical tests to determine the health and integrity of LRU/WRA chassis/backplane conductive paths or single wire/wire harness and weapon system EWIS. Diagnostic equipment shall be capable of detecting and isolating intermittent faults which may occur on multiple conductive paths. For the purposes of this specification, EWIS only refers to metallic electrical interconnections and not fiber optics or pressurized waveguides. Controlled-impedance type cabling (e.g., coaxial/triaxial/high-speed data lines/etc.) may be tested for intermittent shorts and opens but is not required to be tested for complex characteristics (e.g., impedance, insertion/return loss, standing wave ratio, etc.).

3.10.1 Intermittent fault detection (see 4.5.3.1). The diagnostic equipment shall detect intermittent faults of the applicable category as defined in 1.2, as well as isolate the fault to the specific conductive path(s) (see 3.10.1.1 and 6.5.8). In addition, the diagnostic equipment shall perform the following fault identification functions unless otherwise specified in the applicable specification sheet.

3.10.1.1 Intermittent fault isolation (see 4.5.3.1.1). The diagnostic equipment shall identify the LRU/WRA chassis/backplane or single wire/wire harness and weapon system EWIS conductive paths where the intermittent fault occurred and identify as intermittent shorts or intermittent opens as appropriate. At a minimum, the diagnostic equipment shall identify all affected wires/conductive paths. The diagnostic equipment shall utilize the reference designation/nomenclatures of the UUT (e.g., "J1-2", "P1-B") when displaying fault information to the user. Isolation may also include distance to fault, duration, and frequency of the intermittent fault or other additional information as specified in the applicable specification sheet to aid in isolating the fault (see 6.5.8).

3.10.1.2 Continuity (see 4.5.3.1.2). The diagnostic equipment shall be capable of programmable continuity checks against referenced values. For example, the diagnostic equipment should be able to measure conductive path continuity and compare to continuity values from manually entered data or known good LRU/WRA or weapon system EWIS. Continuity resistance (Ohms) measurements shall be accurate to ± 0.5% of the range ± 0.2 ohm.

3.10.1.3 Shorts (see 4.5.3.1.3). The diagnostic equipment shall be capable of detecting and isolating short circuits and tracing/displaying the affected conductive path (see 6.5.16). The equipment shall be capable of detecting short circuit resistance values of 0 ohms to 1 megaohm (1,000,000 ohms) or greater.

3.10.1.4 Conductive path mapping (see 4.5.3.2). Diagnostic equipment shall be capable of mapping conductive paths and generating a map of circuits (schematic) identifying the "as-wired" circuit configuration of the LRU/ WRA (see 6.5.12 and 6.5.19) chassis and backplane circuits and their wire harnesses; weapon system EWIS (see 6.5.4 and 6.5.20); and patch cables, electronic test

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cables, and their connectors. The output of the conductive path mapping shall be recorded in a common machine-readable format (e.g., comma separated variable (csv) files and shall be readily transferrable from the tester using an approved means of data transfer.

3.10.1.5 Data logging (see 4.5.3.3). The diagnostic equipment shall be capable of automatically logging all test results for analysis purposes. At a minimum, logged data shall include the tested path and expected/actual values for both passing and failing measurements. Log files shall also include information on the number/type of faults, as well as identifying the conductive path and all other information presented to the user to aid in fault identification. The log files shall also include information relevant to the LRU/WRA or weapon system EWIS, including identifying characteristics such as UUT part number/nomenclature, date/time of test event, and LRU/WRA or weapon system EWIS serial number/weapon system identification information as appropriate. Data shall be recorded in a common machine-readable format (e.g., csv files) and shall be readily transferrable from the tester using an approved means of data transfer.

3.10.2 False positive/false alarm detection rate (see 4.5.3.4). Following intermittence detection sensitivity adjustment (see 3.10.2.1), the diagnostic equipment's intermittent fault false positive/false alarm detection rate (see 6.5.7) shall be less than one percent. The diagnostic equipment shall detect a minimum of 95 percent of all intermittent faults.

3.10.2.1 Sensitivity. The diagnostic equipment shall have the capability to adjust the intermittence detection sensitivity to eliminate background noise and external factors affecting intermittent fault detection performance. The sensitivity of the diagnostic equipment may be adjusted during intermittence testing so that no extraneous noise is detected. The sensitivity adjustment shall not reduce fault detection performance.

3.10.3 Integrity of point-to-point (pin-to-pin) wiring measurements (see 4.5.3.5). The diagnostic equipment shall display test program parameters and output data (see 3.10.1.5) to indicate the number of occurrences and identify the conductive path(s) in which intermittent faults have occurred.

3.10.4 Stimulus/energization output (see 4.5.3.6). The maximum stimulus voltage shall be ±15 volts, maximum current of 100 milliamperes, and peak current of 200 milliamperes pulsed at 1 millisecond with a 10% duty cycle. When the voltage and current are applied during the intermittent diagnostic testing, the voltage and current provided shall not damage or degrade the LRU/WRA, weapon system EWIS, or patch cables, electronic test cables, and their connectors. In addition to the stimulus voltage used to detect intermittent faults, diagnostic equipment shall be capable of outputting 28 VDC and up to 3 amperes for the purpose of energizing relays and/or other active components that have multiple wiring configurations (see 6.1.3). The diagnostic equipment shall control the energization utilizing its test program software.

3.10.5 Diagnostic equipment start-up and power-on self-test (see 4.5.3.7). When energized or restarted, the diagnostic equipment shall run all power-on self-tests and be ready to operate in a typical work center in less than ten minutes. The diagnostic equipment shall provide automatic diagnostic information indicating whether it is operating within performance requirements. If the equipment is outside the limits of the performance specifications, the information provided by the self-test shall identify the likely cause of the fault.

3.10.5.1 Internal loopback test (see 4.5.3.7.1). The diagnostic equipment shall conduct a loopback test of the diagnostic equipment input/output connections. The purpose of the loopback test is to ensure the integrity of the wiring between internal measurement equipment and interface

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connectors of the diagnostic equipment and does not include the interface cabling required to connect the equipment to the UUT. The time to conduct the internal loopback test shall be specified in the applicable specification sheet and is separate and in addition to the start-up and power-on self-test.

3.10.5.2 Test adapter cable loopback test (see 4.5.3.7.2). The diagnostic equipment shall have the capability to conduct a loopback test between the diagnostic equipment input/output connections and the end connectors of the test adapter cable connector/s of the applicable test program set, provided the interface cables are configured with loopback shorting plugs. Loopback shorting plugs are employed to verify the interface cable integrity. The purpose of this loopback test is to ensure the inter-connection test adapter cable between the diagnostic equipment and the UUT does not have any electrical defects and is configured correctly. The time to conduct the loopback test shall be specified in the applicable specification sheet and is separate and in addition to the start-up and power-on self-test (see 6.1.1).

3.10.6 Operating system. The operating system shall be an industry standard employing a graphical user interface. Updates to the operating system and the application software shall be easily installed without modification to the diagnostic equipment.

3.10.7 Information Technology (IT) compatibility (see 4.6). Diagnostic equipment manufactured in accordance with this specification shall continue to securely operate without compromised confidentiality, integrity, or availability in a defined cybersecurity threat.

3.10.8 Data transfer. The diagnostic equipment shall have the ability to transfer test data. The data transfer capability shall be implemented in accordance with applicable DoD/service-specific cybersecurity regulations as defined by the procuring activity (see 6.1.2).

3.10.9 Test results validity. The diagnostic equipment shall provide repeatable, reliable test results. Results shall not be affected by external factors (e.g., electromagnetic interference (EMI) radiated from external sources). With the same test parameters, test results shall be unchanged after three re-tests.

3.11 Hazardous materials, ozone-depleting substances, and hazardous air pollutants.

3.11.1 Ozone-depleting substances (ODSs). Title VI, Section 606 of the Clean Air Act calls for the elimination of the production of Class I ODSs by December 1995 and Class II ODSs by 2030 (with a 65 percent reduction in the production of Class II ODSs by 2010). No Class I ODS (as defined in Title VI of the Clean Air Act) or material containing a Class I ODS as an ingredient will be approved for use during any phase of the system's life cycle, which includes manufacture, operation, maintenance, and disposal.

3.11.2 Hazardous materials. The diagnostic equipment shall not require the use of hazardous or environmentally unacceptable materials throughout its life cycle unless there is no feasible alternative. Hazardous materials are those meeting one or more of the following conditions:

a. Regulated as a hazardous material in accordance with 49 CFR 173.

b. Requires a Safety Data Sheet (SDS) in accordance with 29 CFR 1910.1200.

c. Regulated as an Extremely Hazardous Substance (EHS) in accordance with 40 CFR 355, Appendices A and B.

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d. Regulated as a Toxic Chemical in accordance with 40 CFR 372.65.

e. Meets or has the potential to meet the definition of hazardous waste, as defined by 40 CFR 261 Subparts A, B, C, or D, during end-use, treatment, handling, packaging, storage, transportation, or disposal.

f. Regulated as an Ozone Depleting Substance (ODS) in accordance with 40 CFR 82 Subpart A, Appendices A and B.

g. Identified in the Clean Air Act, Chapter 85, Subchapter I - 7412 as a Hazardous Air Pollutant (HAP).

3.12 Environmental characteristics. The diagnostic equipment shall meet the performance characteristics specified herein (see 3.10) when subjected to the following operating/non-operational environmental requirements unless otherwise specified in the applicable specification sheet.

3.12.1 Depot-level diagnostic equipment. Depot-level diagnostic equipment is designed for benign operational environments where the environmental conditions are controlled and protected. Controlled and protected areas include a service area such as a laboratory. See MIL-PRF-28800, Class 4, for more information regarding a depot-type environment. The diagnostic equipment shall be capable of operating under the following conditions unless otherwise specified in the applicable specification sheet.

a. Temperature: 77 °F ±18 °F (25 °C ±10 °C).

b. Humidity: 20 to 70 percent relative humidity.

c. Altitude: 0 to 15,092 feet (4,600 meters).

3.12.2 Field-level diagnostic equipment. Field level diagnostic equipment is designed for rugged operational environments where the environmental conditions are routinely encountered in an unprotected, uncontrolled climate. Portable diagnostic equipment designed for use in more than one location with a substantial degree of environmental protection shall be considered field level. See MIL-PRF-28800, Class 2, for more information regarding field-level diagnostic equipment. The diagnostic equipment shall be capable of operating under the following conditions, unless otherwise specified in the applicable specification sheet.

3.12.2.1 Temperature, not operating (see 4.5.5.2.1). The diagnostic equipment shall meet the performance characteristics of 3.10 after storage at temperatures between -60 to 160 °F (-51 to 71 °C).

3.12.2.2 Temperature, operating (see 4.5.5.2.1). The diagnostic equipment shall meet the performance characteristics of 3.10 at operating temperatures between -14 to 131 °F (-40 to 55 °C) and 20 minute operation at 160 °F (71 °C).

3.12.2.3 Humidity (see 4.5.5.2.1). The diagnostic equipment shall meet the performance characteristics of 3.10 after exposure to humidities of 5 to 75 ±5 percent at 86 to 104 °F (30 to 40 °C) and 5 to 45 ±5 percent above 104 °F (40 °C).

3.12.2.4 Altitude (see 4.5.5.2.2). The diagnostic equipment shall meet the performance characteristics of 3.10 when operated at an altitude of 15092 feet (4600 meters) and after returning from an altitude of 15091 feet (4600 meters), when tested in accordance with 4.5.5.2.2.

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3.12.2.5 Fungus resistance (see 4.5.5.2.3). The diagnostic equipment shall not contain materials that provide nutrients for the growth of fungus.

3.12.2.6 Random vibration (see 4.5.5.2.4). The diagnostic equipment shall meet the performance characteristics of 3.10 after completing random vibration testing.

3.12.2.7 Bench handling (see 4.5.5.2.5). The diagnostic equipment shall meet the performance characteristics of 3.10, and there shall be no damage to controls, indicators, or fuse holders after being tested in accordance with 4.5.5.2.5.

3.12.2.8 Electromagnetic compatibility (EMC) (see 4.5.5.2.6). The diagnostic equipment shall meet the performance characteristics of 3.10 during electromagnetic compatibility testing in accordance with MIL-STD-461: Ground, Navy, Army, and Air Force.

3.12.2.9 Loose cargo bounce (see 4.5.5.2.7). The diagnostic equipment shall meet the performance characteristics of 3.10, and there shall be no damage to controls, indicators, or fuse holders after being tested in accordance with 4.5.5.2.7.

3.13 Marking and identification. Diagnostic equipment shall be marked with appropriate identification in accordance with MIL-STD-130.

3.14 Reliability. The diagnostic equipment shall have a mean time between failure in excess of 1500 hours.

3.15 Maintainability. The diagnostic equipment shall meet the maintainability requirements of MIL-PRF-28800.

3.16 Special tools and equipment. Special tools required to maintain and operate the SE shall be provided. If additional equipment to support the diagnostic equipment is required, items already in the Government's inventory shall be used whenever practicable.

3.17 Workmanship. The diagnostic equipment shall be free from irregularities or defects that could degrade performance or durability.

3.18 Safety. The diagnostic equipment shall meet the safety requirements of MIL-PRF-28800.

3.19 Government validation. The diagnostic equipment may be tested in a laboratory environment by the Government using an Intermittent Fault Emulator (IFE) (see 4.8) at the discretion of the procuring activity. The IFE is intended to emulate LRU/WRA or weapon system EWIS intermittent faults to evaluate intermittent fault diagnostic equipment. LRU/WRA or weapon system EWIS intermittent faults may occur at random times, maybe of varying durations, and may occur on single or multiple LRU/WRA (see 6.5.12 and 6.5.19) chassis and backplane circuits and their wire harnesses; weapon system EWIS (see 6.5.4); and patch cables, electronic test cables and their connectors. An LRU/WRA cannot be used to evaluate the diagnostic equipment capabilities because of these random characteristics. The intermittent fault is emulated by the IFE by opening and closing electronic switches for known times, durations, and conductive path locations. By these means, the diagnostic equipment's intermittent fault detection can be compared to the emulated intermittent fault(s) created by the IFE. See MIL-HDBK-527 for guidance on the use and application of the IFE.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified herein.

4.3 First article. First article inspections shall be performed on diagnostic units when required in accordance with the table I testing requirements for depot level equipment and in accordance with table II for field-level equipment. The number of first article units shall be as required in the contract or purchase order (see 6.2).

TABLE I. First article inspections for depot level diagnostic equipment (MIL-PRF-28800, Class 4).

Inspection title (Examination or test)	Requirement paragraph	Test method paragraph
Mechanical and visual examination	3.6, 3.8.1, 3.8.2, 3.9.1, 3.9.2, 3.10.6, 3.10.8	4.4.1
Information Technology (IT). Compatibility	3.10.7	4.6
Input power	3.9.3	4.7.1
Interruption of power source	3.9.3.1	4.7.2 and 4.7.3
Intermittent fault detection	3.10.1	4.5.3.1
Intermittent fault isolation	3.10.1.1	4.5.3.1.1
Continuity	3.10.1.2	4.5.3.1.2
Shorts	3.10.1.3	4.5.3.1.3
Conductive path mapping	3.10.1.4	4.5.3.2
Data logging	3.10.1.5	4.5.3.3
False positive/false alarm detection	3.10.2	4.5.3.4

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TABLE I. First article inspections for depot level diagnostic equipment (MIL-PRF-28800, Class 4) - Continued.

Inspection title (Examination or test)	Requirement paragraph	Test method paragraph
Integrity of point-to-point (pin-to-pin) wiring measurements	3.10.3	4.5.3.5
Stimulus/energization output	3.10.4	4.5.3.6
Diagnostic equipment start-up and power-on self-test	3.10.5	4.5.3.7
Internal loopback test	3.10.5.1	4.5.3.7.1
Test adapter cable loopback test	3.10.5.2	4.5.3.7.2
Operational test	---	4.5.4

TABLE II. First article inspections for field-level diagnostic equipment (MIL-PRF-28800, Class 2).

Inspection title (Examination or test)	Requirement paragraph	Test method paragraph
Mechanical and visual examination	3.6, 3.8.1, 3.8.2, 3.9.1, 3.9.2, 3.10.6, 3.10.8	4.4.1
Information Technology (IT) compatibility	3.10.7	4.6
Input power	3.9.3	4.7.1
Interruption of power source	3.9.3.1	4.7.2, 4.7.3
Intermittent fault detection	3.10.1	4.5.3.1
Intermittent fault isolation	3.10.1.1	4.5.3.1.1
Continuity	3.10.1.2	4.5.3.1.2
Shorts	3.10.1.3	4.5.3.1.3
Conductive path mapping	3.10.1.4	4.5.3.2
Data logging	3.10.1.5	4.5.3.3
False positive/false alarm detection rate	3.10.2	4.5.3.4
Integrity of point-to-point (pin-to-pin) wiring measurements	3.10.3	4.5.3.5
Stimulus/energization output	3.10.4	4.5.3.6
Diagnostic equipment start-up and power-on self-test	3.10.5	4.5.3.7
Internal loopback test	3.10.5.1	4.5.3.7.1
Test adapter cable loopback test	3.10.5.2	4.5.3.7.2

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TABLE II. First article inspections for field-level diagnostic equipment (MIL-PRF-28800, Class 2) - Continued.

Inspection title (Examination or test)	Requirement paragraph	Test method paragraph
Temperature, not operating	3.12.2.1	4.5.5.2.1
Temperature, operating	3.12.2.2	4.5.5.2.1
Humidity	3.12.2.3	4.5.5.2.1
Altitude	3.12.2.4	4.5.5.2.2
Fungus resistance	3.12.2.5	4.5.5.2.3
Random vibration	3.12.2.6	4.5.5.2.4
Bench handling	3.12.2.7	4.5.5.2.5
Electromagnetic compatibility	3.12.2.8	4.5.5.2.6
Loose cargo bounce	3.12.2.9	4.5.5.2.7

4.4 Conformance inspection. The conformance inspections shall include the following inspections and tests:

4.4.1 Mechanical and visual examination. The equipment shall be given a thorough mechanical and visual examination and test to determine that all materials, workmanship, and safety characteristics comply with the specified requirements.

4.4.2 Electrical circuit configuration. The equipment shall be examined or tested to confirm that the wiring is correct. Where applicable, the tests shall include the requirements specified in a and b below:

a. All intra-module wiring shall be tested to ensure correctness.

b. The module grounding system shall be examined or tested to ensure proper separation of shield, signal, and framework grounds, and metal-to-metal contact for panels and components that serve as electromagnetic shields.

4.5 Test methods.

4.5.1 Test conditions. Unless otherwise specified in the detailed test herein, the inspection of 4.5 shall be performed under the conditions of a through d. Ambient conditions within the specified ranges need not be controlled. For equipment having a specified warmup or stabilization period after turn-on, formal measurements and observations shall be made only after the specified interval (see 3.8.1.1).

a. 77 °F ±18 °F (25 °C ±10 °C)

b. Humidity: 20 to 70 percent relative humidity (RH)

c. Altitude: 0 to 15,092 feet (4,600 meters).

d. Power: Nominal or alternate power source specified for the equipment.

4.5.2 Mechanical and visual examination (see 3.6, 3.8, 3.9, and 3.10). The equipment shall be given a thorough mechanical and visual examination and test to determine that all materials, workmanship, and safety characteristics comply with the specified requirements. The equipment shall be examined for the following:

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a. User interface (see 3.8.1). The operation of the diagnostic equipment shall be demonstrated to validate the use of commercially available non-proprietary software.

b. Expandability (see 3.8.2). The number of test points and connections for modular expansion units as specified by the procuring activity. The operation of the diagnostic equipment shall be demonstrated to validate that testing is controlled by a central control unit.

c. Size and weight (see 3.9.1). The size and weight of the diagnostic equipment shall be measured for compliance to the weight and size limits specified by the procuring activity.

d. Interface cables (see 3.9.2). The diagnostic equipment shall be examined for interface cable connectors as specified by the procuring activity. Examine diagnostic equipment connections to verify the equipment interfaces with existing interface cables identified by the purchaser in the acquisition requirements (see 6.2).

e. Operating system. The operating system and means for updating software shall be verified for compliance to 3.10.6.

f. Data transfer. The means for transferring test data (e.g., test results, wire signatures, and test programs) to an external device shall be verified for compliance to 3.10.8.

4.5.3 Performance characteristic testing. The diagnostic equipment shall be installed in the test facility or location in a manner that will simulate service usage, making connections, and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be covered. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the test item is installed.

4.5.3.1 Intermittent fault detection verification (see 3.10.1). Unless otherwise specified in the specification sheet, fault detection performance characteristics of the diagnostic equipment shall be verified using the IFE (see MIL-HDBK-527). Perform the operational test in accordance with 4.5.4. If a performance check is required during operation, perform 4.5.4. Once testing is concluded, verify diagnostic equipment performance, and perform the post-test inspection in accordance with 4.5.6.

4.5.3.1.1 Intermittent fault isolation (see 3.10.1.1). When configured in accordance with 4.5.3, the diagnostic equipment shall isolate and identify over 95% of intermittent fault circuit(s) in the UUT or IFE, in keeping with the category and capability of the equipment (see 1.2). A minimum of 50 conductive paths shall be tested in the UUT and/or IFE.

4.5.3.1.2 Continuity (see 3.10.1.2). When configured in accordance with 4.5.3, the diagnostic equipment shall isolate and identify all open circuit faults in the UUT or IFE in keeping with the category and capability of the equipment (see 1.2). A minimum of 50 conductive paths shall be tested in the UUT or IFE.

4.5.3.1.3 Shorts (see 3.10.1.3). When configured in accordance with 4.5.3, the diagnostic equipment shall isolate and identify all short circuit faults in the UUT or IFE, in keeping with the category and capability of the equipment (see 1.2). A minimum of 50 conductive paths shall be tested in the UUT or IFE.

4.5.3.2 Conductive path mapping (see 3.10.1.4). When configured in accordance with 4.5.3,

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the diagnostic equipment shall display the mapped conductive paths of the UUT or IFE and output of the results in a common machine-readable format (e.g., csv files) from the tester using approved means of data transfer.

4.5.3.3 Data logging (see 3.10.1.5). When configured in accordance with 4.5.3, the diagnostic equipment shall display all test results for analysis purposes. Data shall be recorded and output in a common machine-readable format (e.g., csv files) and shall be readily transferrable from the tester using approved means of data transfer.

4.5.3.4 False positive/false alarm detection rate (see 3.10.2). When configured in accordance with 4.5.3, the diagnostic equipment shall isolate and identify over 95 percent of intermittent circuit faults in the UUT or IFE, in keeping with the category and capability of the equipment (see 1.2). The UUT tested shall contain a minimum of 50 conductive paths capable of testing.

4.5.3.5 Integrity of point-to-point (pin-to-pin) wiring measurements (see 3.10.3). When configured in accordance with 4.5.3, the diagnostic equipment shall verify and test program parameters and output the data using an approved means of data transfer. The displayed data results shall indicate the number of occurrences and identify the conductive path(s) in which intermittent faults have occurred in a minimum of ten consecutive attempts, with 100% accuracy. A minimum of 50 conductive paths shall be tested in the UUT or IFE.

4.5.3.6 Stimulus/energization output. Stimulus voltage and current of the diagnostic equipment shall be measured for compliance to 3.10.4.

4.5.3.7 Diagnostic equipment start-up and power-on self-test (see 3.10.5). When configured in accordance with 4.5.3, the diagnostic equipment shall perform a power-on self-test and be functional within 10 minutes. Upon power-up, the equipment shall provide the user with its operational status. If the equipment is outside the limits of the performance specifications, the information provided by the self-test shall identify the likely cause of the fault.

4.5.3.7.1 Internal loopback test (see 3.10.5.1). When configured in accordance with 4.5.3, the diagnostic equipment shall perform a loopback test of the diagnostic equipment input/output connections. The capability of the diagnostic equipment to perform a loopback test of the diagnostic equipment input/output connections shall be demonstrated. The loopback test shall successfully detect all open/short circuit faults induced during the demonstration and isolate them to the connector and contact(s) that failed the loopback test. A minimum of 1 open and 1 short shall be induced at random during the loopback test. The loopback test may include the insertion of a temporary, in-line test adapter using a jumper or a shunt.

4.5.3.7.2 Test adapter cable loopback test (see 3.10.5.2). The diagnostic equipment shall conduct a loopback test between the diagnostic equipment input/output connections and the end connectors of the test adapter cable connector(s) of the applicable test program set. The capability of the diagnostic equipment to perform a loopback test between the diagnostic equipment input/output connections and the end connectors of the test adapter cable connector/s of the applicable test program set shall be demonstrated. The loopback test shall successfully detect all open/short circuit faults induced within the adapter cable during the demonstration and isolate them to the connector and contact(s) that failed the loopback test. A minimum of 1 open and 1 short shall be induced at random during the loopback test. The loopback test shall utilize a typical test adapter cable set that incorporates shorting/loopback/self-test plugs to verify the test adapter cable integrity.

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4.5.4 Operational test. A multi-channel detection test shall be conducted to test the operational performance of the diagnostic equipment using the IFE or equivalent. Using the IFE or equivalent, perform the sequence of tests listed in table III. Perform test 1-1, and if intermittent faults are detected on at least 95 percent of the test points, then proceed to test 1-2 through test 1-6. If the Intermittent Fault Diagnostic Equipment passes all 6 tests, then it can be concluded that the diagnostic equipment is able to test across multiple test points. In addition, these tests demonstrate that the diagnostic equipment can detect intermittent faults down to 25 microseconds (40 kHz), which is within the Category 2 range.

TABLE III. Multi-channel detection tests.

Test number	Number of conductive paths evaluated	MIL-PRF-32516 category	Fault duration
1-1	128	Not categorized	500 milliseconds
1-2	128	3	5 milliseconds
1-3	256	3	10 milliseconds
1-4	256	3	501 microseconds
1-5	256	2	250 microseconds
1-6	256	2	25 microseconds

4.5.4.1 Extended test procedure (category 1 only). After successfully completing the tests of 4.5.3, the following tests determine if the Intermittent Fault Diagnostic Equipment can detect intermittent faults less than 100 nanoseconds (10 MHz) in duration.

4.5.4.1.1 Signal generator (SG). This testing requires the use of an SG with the ability to generate an arbitrary waveform generator output. Set the SG with a square wave with peak to peak voltage set to 1.5 volts and the frequency set as outlined in table IV, and place the selected test points across the SGs output. The circuit will produce a raw simulated intermittent event each time the SG is triggered, and the operator can assess if the diagnostic equipment has detected the simulated fault.

4.5.4.1.2 Extended tests. This testing should include the tests listed in table IV as a minimum. Note: For ease of setup and operation, it may be more appropriate to use one test point on the Intermittent Fault Diagnostic Equipment and carry out each of the frequencies three times for each test point.

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TABLE IV. Extended tests.

Test number	Method	Category	SG fault duration
2-1	Use 5 different test points on the diagnostic equipment and trigger the SG 3 times with a gap of 1 second between triggers. Record how many of the 15 faults are detected by the diagnostic equipment.	2	25 microseconds (40 kHz)
2-2		2	5 microseconds (200 kHz)
2-3		2	1 microsecond (1 MHz)
2-4		2	500 nanoseconds (2 MHz)
2-5		2	125 nanoseconds (8 MHz)
2-6		1	95 nanoseconds (10.5 MHz)

4.5.5 Environmental testing.

4.5.5.1 Depot level testing. The diagnostic equipment shall be installed in the test facility or location in a manner that will simulate service usage, making connections, and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be covered. The diagnostic equipment shall be operated to determine that no malfunction or damage was caused due to faulty installation or handling. Examination and testing shall include mechanical and visual examination (see 4.5.2); performance characteristic testing (see 4.5.3); and operational test (see 4.5.4).

4.5.5.2 Field level testing. Examination and testing shall include the following unless otherwise specified in the applicable specification sheet.

4.5.5.2.1 Temperature and humidity (see 3.12.2.1, 3.12.2.2, 3.12.2.3). Temperature and humidity tests shall be conducted in accordance with MIL-PRF-28800, Class 2. There shall be no evidence of rust or corrosive contaminants following the completion of testing. Performance of the operational test shall occur at the end of each temperature test period, adding whatever time is required to perform the satisfactory operational test. This means that the total time required to perform the temperature testing will be the cumulative total consisting of the time required for each temperature test, the time required to perform a satisfactory operational test at each temperature test, and any interruption period.

a. Temperature test:

Test (a): Operating - 2 hours at room ambient (see 4.5.1, temperature);

Test (b): Non-Operating 4 hours at -14 °F (-10 °C);

Test (c): Non-Operating 4 hours at - 60 °F (-51 °C), followed by Non-Operating - 4 hours at room ambient;

Test (d): Operating 4 hours at 131 °F (55 °C), and humidity controlled at 5 to 20%;

Test (e): Non-Operating - 4 hours at 71 °C, followed by Non-Operating - 4 hours

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at room ambient.

b. Humidity test:

Test (a): 45% Relative humidity at 131 °F (55 °C);

Test (b): 75% Relative humidity at 104 °F (40 °C);

Test (c): 95% Relative humidity at 86 °F (30 °C).

4.5.5.2.2 Altitude test (see 3.12.2.4). The altitude test shall be conducted in accordance with MIL-PRF-28800, Class 2. The operational test (see 4.5.4) shall be conducted to determine satisfactory performance in accordance with the MIL-PRF-28800 altitude test.

4.5.5.2.3 Fungus resistance (see 3.12.2.5). The manufacturer shall provide a certified statement stating that no organic material is used in the manufacturing of the diagnostic equipment.

4.5.5.2.4 Random vibration tests (see 3.12.2.6). Random vibration testing shall be conducted in accordance with MIL-PRF-28800, Class 2. At the conclusion of the vibration test, conduct a physical evaluation and operational test (see 4.5.4).

4.5.5.2.5 Bench handling test (see 3.12.2.7). Bench handling test shall be conducted in accordance with MIL-PRF-28800, Class 2. The operational test (see 4.5.4) shall be conducted during testing to determine satisfactory performance.

4.5.5.2.6 Electromagnetic compatibility (EMC) (see 3.12.2.8). The diagnostic equipment shall be tested in accordance with MIL-STD-461 specific requirements of CE102, CS101, CS114, CS115, CS116, RE102, and RS103.

4.5.5.2.7 Loose cargo bounce (see 3.12.2.9). Loose cargo bounce test shall be conducted in accordance with MIL-PRF-28800. Figure 1 is an example of the test bed used for the loose cargo bounce test. At the conclusion of the test, conduct a physical evaluation and operational test (see 4.5.3).

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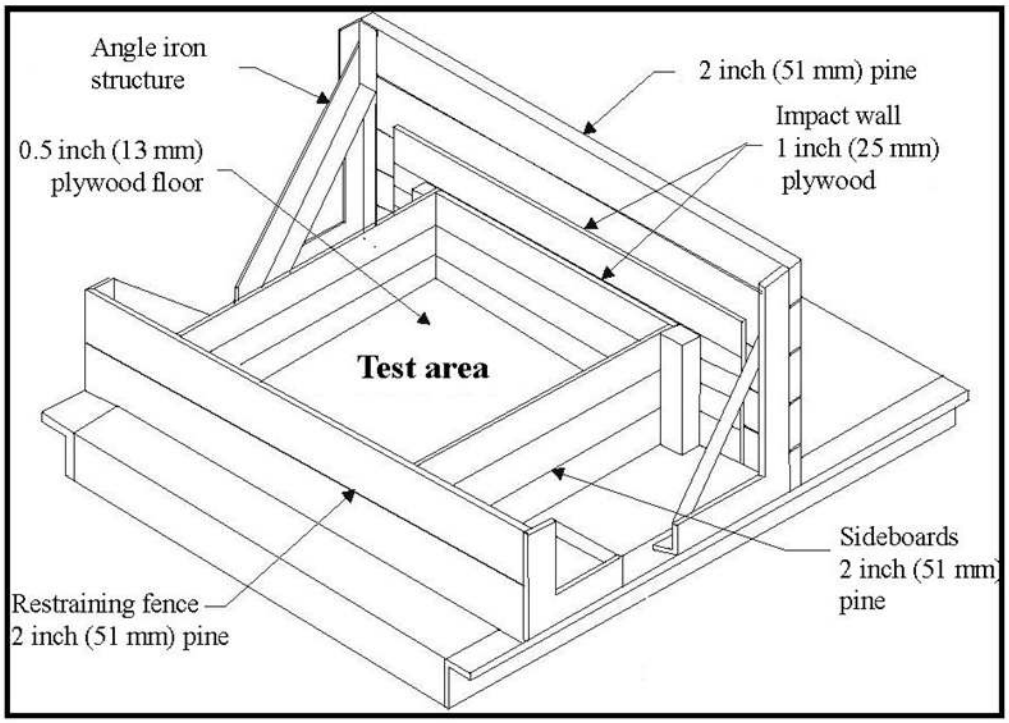


FIGURE 1. Package tester fixture example (see 4.5.5.2.7).

4.5.6 Post-test inspection. At the completion of each environmental test, the diagnostic equipment shall be inspected in accordance with the pretest (see 4.5.4). The diagnostic equipment shall have failed the test when any of the conditions specified in a through g occur:

a. Monitored functional parameters deviate beyond acceptable limits established in 4.5.4.

b. Catastrophic or structural failure.

c. Mechanical binding or loose parts, including screws, clamps, bolts, and nuts, that result in component failure or a hazard to personnel safety.

d. Malfunction.

e. Degradation of performance beyond limits established in the specification sheet.

f. Any additional deviations from acceptable criteria established before the test.

g. Deterioration, corrosion, or change in tolerance limits of any internal or external parts that could in any manner prevent the test item from conforming to operational service or maintenance requirements.

4.6 Information Technology (IT) compatibility (see 3.10.7). The diagnostic equipment shall be tested and comply with DoD cybersecurity policies as defined in DoDI 8500.01, which will allow the system to receive authorization to operate (ATO) based upon the Risk Management Framework (RMF) process as defined in DoDI 8510.01. The diagnostic equipment shall be designed to securely operate without compromised confidentiality, integrity, or availability.

4.7 Electrical power tests (see 3.10.5). During the electrical power tests, the equipment performance shall be monitored as specified in 4.7.1 and 4.7.2.

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4.7.1 Input power test (see 3.9.3). The input power test shall be performed by measuring the power consumption of the equipment. Power consumption that exceeds the value specified in the applicable specification sheet shall constitute a failure of this test.

4.7.2 Voltage and frequency variation test (see 3.9.3.1). The response of the equipment to input voltage and frequency variations shall be tested as specified in Steps 1 through 4 at 120 Vrms or 240 Vrms as specified in the applicable specification sheet.

Step 1. Connect the power cable of the equipment to a variable voltage and frequency source.

Step 2. Set the output of the variable source at 120 Vrms or 240 Vrms, 60 Hz.

Step 3. Performance of the equipment outside the limits required in the satisfactory operation check or other tests specified in the applicable specification sheet shall constitute a failure of this test.

Step 4. Repeat Steps 2 and 3 for voltages and frequencies required by the applicable specification sheet from the variable source as specified in table V, using the maximum voltage and minimum voltage limit values. The voltage and frequency shall be maintained for at least 15 minutes at each level specified.

This test may be performed at other points within the specified ranges at the discretion of the Government quality assurance representative.

4.7.2.1 Voltage-transient test. The equipment performance during voltage transients shall be tested as specified in Steps 1 through 7.

Step 1. Connect the power cable of the equipment to a 60 Hz variable voltage source.

Step 2. Set the output of the source to the maximum voltage as specified in table V.

TABLE V. Variable frequency power supply outputs.

Frequency (Hz)	120 Vrms (Nominal)		240 Vrms (Nominal)	
	Min voltage (Vrms)	Max voltage (Vrms)	Min voltage (Vrms)	Max voltage (Vrms)
47.5	108	132	216	264
52.5				
57				
63				

Step 3. Quickly increase the source voltage to a transient value of 156 Vrms (120 Vrms nominal) or 288 Vrms (240 Vrms nominal), hold for approximately 0.5 second, and quickly return to the starting voltage of Step 2. Monitor equipment performance during and after the transient for compliance with the satisfactory operation check or other tests specified in the applicable specification sheet. Performance outside of the specified limits for longer than 30 seconds shall constitute a failure of this test.

Step 4. Repeat Step 3 four more times at 2-minute intervals.

Step 5. Set the output of the source to a starting voltage of 108 Vrms (120 Vrms nominal) or 216 Vrms (240 Vrms nominal).

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Step 6. Repeat Step 3 with a transient value of 84 Vrms (120 Vrms nominal) or 192 Vrms (240 Vrms nominal) and the starting voltage specified in Step 5.

Step 7. Repeat Step 6 four more times at 2-minute intervals.

4.7.2.2 Frequency-transient test. The equipment performance during frequency transients shall be tested as specified in Steps 1 through 10.

Step 1. Connect the power cable of the equipment to a 120 Vrms or 240 Vrms variable frequency source. The output voltage shall be the operating voltage specified for the equipment.

Step 2. Set the output of the source to a starting frequency of 47.5 Hz.

Step 3. Quickly change the source frequency to a transient value of 45 Hz, hold for approximately 0.5 second, and quickly return to the starting frequency of 47.5 Hz. Monitor equipment performance during and after the transient for compliance to the satisfactory operation check or other tests specified in the applicable specification sheet. Performance outside of the specified limits for longer than 30 seconds shall constitute a failure of this test.

Step 4. Repeat Step 3 four more times at 1-minute intervals.

Step 5. Set the output frequency of the source to a starting frequency of 57 Hz.

Step 6. Repeat Step 3 with a transient frequency of 54 Hz and a starting frequency of 57 Hz.

Step 7. Repeat Step 6 four more times at 1-minute intervals.

Step 8. Set the output frequency of the source to a starting frequency of 63 Hz.

Step 9. Repeat Step 3 with a transient frequency of 66 Hz and a starting frequency of 63 Hz.

Step 10. Repeat Step 9 four more times at 1-minute intervals.

4.7.3 Power source interruption test (see 3.9.3.1). The power source interruption test shall conform to the requirements of 3.9.3.1.

4.8 Government validation (see 3.19). The IFE shall be used to validate diagnostic equipment capability to detect and isolate intermittent faults from 100 nanoseconds to 500 milliseconds (see MIL-HDBK-527). The IFE emulates intermittent changes in impedance within a UUT as the UUT is subjected to external stimuli (e.g., temperature and/or vibration). The procedures identified in MIL-HDBK-527 are a two-step process. As recommended by MIL-HDBK-527, the first step is to evaluate the multi-channel capability of diagnostic equipment using the IFE. The second step is to evaluate the diagnostic equipment using a signal generator to determine the equipment's capability to detect events down to 100 nanoseconds. The IFE is capable of generating multiple intermittent faults to simulate 256 LRU/WRA conductive paths. The emulator is capable of generating controlled intermittent faults of various durations from 100 nanoseconds to 500 milliseconds at pseudo-random time intervals. The diagnostic equipment shall detect a minimum of 95 percent of the emulated intermittent faults (see 3.10.2). The use of additional

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equipment to validate the diagnostic equipment's ability to detect and isolate intermittent faults shall be approved by the procuring activity. See MIL-HDBK-527 for guidance on the use and application of the IFE.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The diagnostic equipment defined in this specification is intended for use in detecting and isolating intermittent faults in LRUs/WRAs (see 6.5.12 and 6.5.19) chassis and backplane circuits and their wire harnesses; weapon system EWIS (see 6.5.4); and patch cables, electronic test cables and their connectors. The diagnostic equipment may be used on weapon system EWIS or with the EWIS component or LRU/WRA removed from the weapon system. When testing is performed off-weapon system, on LRUs/WRAs, the SRAs/SRUs are typically removed. The off-weapon system units under test (EWIS component or LRU/WRA) are stimulated by temperature, vibration, or vibration/temperature to emulate the operating environment in which the fault originally occurred. When the diagnostic equipment is used to isolate and detect intermittent faults in EWIS components installed on the weapon system, the UUT is isolated and disconnected from other system LRUs/WRAs. This test method is limited in its ability to emulate the aircraft/weapon system's operating environment, which limits the ability to isolate and detect intermittent faults. Appendix A, B, and C provide recommended guidance for defining the simulated operating environment.

6.1.1 Interface cabling. Interface cabling may be cabling specifically designed for the diagnostic equipment or existing ITA, IACS, or TAC (see 3.9.2). This cabling should be tested by the diagnostic equipment for intermittent faults prior to testing LRU/WRA, chassis, and backplanes and their wire harnesses, or weapon system EWIS (see 3.10.5.2).

6.1.2 Computer interface. Diagnostic equipment utilizing any manner of computer interface and/or data storage capability is required to comply with applicable DoD and service cybersecurity policies; for information on DoD-wide cybersecurity policies, refer to DoDI 8500.01. Diagnostic equipment is typically operated on "standalone" computers, i.e., computers that are not connected to DoD networks. Transfer of data to and from these computers must be performed in compliance with all applicable DoD/service cybersecurity policies, as well as any local policies in place where the diagnostic equipment is utilized. The diagnostic equipment will be granted an Authorization to Operate (ATO) after testing for compliance with applicable policies. The ATO will be approved based on the Risk Management Framework process defined in DoDI 8510.01.

6.1.3 Stimulus/energization. In order for the diagnostic equipment to be able to detect an intermittent fault, it must output a stimulus voltage to the LRU/WRA, EWIS, or UUT. The

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stimulus voltage is needed for the diagnostic equipment to detect and/or measure the interruption of current flow in the conductive paths under test. The stimulus voltage must be low enough not to damage the LRU/WRA, EWIS, or UUT. Typically the stimulus voltage is applied to the LRU/WRA with the SRAs/SRUs removed so that only the conductive paths are evaluated and damage to sensitive electronic circuitry is avoided. Similarly, EWIS is typically tested end to end with LRUs/WRAs or other components removed so that only the conductive paths are evaluated. Energization voltage is required to open/close any contactor/relay that is installed in the LRU/WRA, EWIS, or UUT conductive paths under test to permit current flow through conductive paths isolated by the contactor/relay.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable purchase description.

c. Appropriate category (see 1.2).

d. Number of first article samples (see 4.3).

e. Packaging (see 5.1)

f. If required, the specific issue of documents.

g. First article inspection.

h. Production lot, conformance inspection.

i. The quantity of maintenance and calibration aid sets required will be as specified by the procuring activity, including circuit board extenders, special adapters, special tools, and patch cables.

j. Waivers are required for equipment that incorporates restricted materials (see 3.11.2). The restricted materials are prohibited except where such materials are fabricated into completed approved standard parts, or use of the material is approved by the procuring activity.

k. The equipment manufacturer must have a standard commercial quality assurance program. For example, the program could be but is not required to be certified to ISO 9001 or 9002. Other such recognized commercial quality assurance programs are acceptable.

l. Failure criteria.

m. If required, the location of the identification plate on a transit case must be specified.

n. The requirement for nomenclature assignment and the nomenclature to be assigned.

o. The equipment manufacturer should have a standard Electrostatic Discharge Control Program that complies with the requirements of ANSI/ESD S20.20.

p. The quantity of accessories including, but not limited to: power cords, fuses (if required), interface cables, etc.

q. The requirement for technical manuals (see technical manuals as defined in MIL-PRF-28800).

r. Microsoft Windows operating system version or Non-Windows operating system.

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s. The requirement for Government Validation. See MIL-HDBK-527 for guidance on the use and application of the IFE.

t. Marking (3.13).

u. Number of test points in accordance with modular unit and the maximum amount of test point expansion.

v. If required, existing ITA or TPS cabling should be identified for compatibility with the diagnostic equipment.

w. Periodic calibration documentation recommending frequency of calibration, calibration equipment requirements, and calibration instructions.

6.3 First article inspection. When first article inspection is required, the equipment should be first production units. The contracting officer should include specific instructions in the procurement documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.

6.4 Applicability of diagnostic equipment. The diagnostic equipment would typically be used on LRUs/WRAs and weapon system EWIS that have demonstrated low reliability and/or repair history of No Fault Found (NFF) or quasi-NFF repair (e.g., cannot duplicate (CND), re-test OK (RETOK), beyond the capability of maintenance (BCM), disassemble-clean-reassemble (DCR), etc.).

6.5 Definitions.

6.5.1 Bench-top equipment. Bench-top equipment is designed to be used on a fixed bench or table or on a mobile cart. Equipment that exceeds 11 pounds (5 kg) in weight and has no handles, or exceeds 44 pounds (20 kg) with or without a handle, is considered to be bench-top equipment.

6.5.2 Conductive path. An interconnection (e.g., metallic wire/cabling, circuit board traces, etc.) that is intended by design to transfer electrical signals or power between two or more terminal points (e.g., connector contacts, terminal lugs, etc.). For the purposes of this document, conductive paths only include metallic interconnections; fiber optics and pressurized waveguides are explicitly excluded from this definition.

6.5.3 Depot maintenance. Maintenance performed on materiel requiring major overhaul or a complete rebuild of parts, subassemblies, and end items, including the manufacture of parts, modifications, testing, and reclamation. Depot maintenance serves to support lower echelons of maintenance by providing technical assistance and performing maintenance beyond their capability. Depot maintenance provides stocks of serviceable equipment by using more extensive facilities for repair than are available in lower-level maintenance activities.

6.5.4 Electrical Wiring Interconnection System (EWIS). EWIS is defined as any wire, electrical cabling, or fiber optic cabling, or a combination of these items, including terminations, installed in any area of the weapon system for the purpose of transmitting electrical or optical energy, signals, or data between two or more electrical/optical endpoints. For the purposes of this specification, EWIS only refers to electrical interconnections and not fiber optics.

6.5.5 Expandability. The ability of the base equipment to be interfaced with expansion equipment in logical increments to handle a growing number of test points in a capable manner or its ability to be enlarged to accommodate that growth.

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6.5.6 Failure. Equipment failure as used herein is any departure from the required performance or operation outside of the required accuracies (not correctable by normal use of the operating controls) or deviation from the criteria of section 3 (see requirements and test methods listed in tables I and II) after the test is initiated.

6.5.7 False positive/false alarm rate. The false positive/false alarm rate is any fault indication by the diagnostic equipment that results from external influences but does not represent any defect in the LRU/WRA or weapon system EWIS under test. An example of an external influence that may trigger an erroneous indication is electromagnetic emissions imparted onto the LRU/WRA or weapon EWIS under test from an external source (e.g., an electric motor or DC switching power supply). For field-level equipment that is working in "dirty" electromagnetic environments, false positives that arise from external influences are of particular concern as they may not represent functional failures within the EWIS under test. This may trigger unnecessary maintenance/troubleshooting actions.

6.5.8 Fault detection/isolation. The ability to isolate the fault to a single conductive path (multiple paths in the case of a short circuit or intermittent fault) during the evaluation of single or multiple conductive paths. Isolation may also include distance to fault, duration, and frequency of the intermittent fault or other additional information as specified in the applicable specification sheet to aid in isolating the fault.

6.5.9 Intermediate duration fault. An intermediate duration fault is a fault which has a duration between 101 nanoseconds and 500 microseconds and typically has a resistance between 10 to 500 ohms.

6.5.10 Intermittent faults. Intermittent faults are short-duration impedance variations (opens/shorts) that occur in conductive paths in LRU/WRA chassis/backplanes or weapon system EWIS. Intermittent faults occur as a result of various operational environmental stimuli, including, but not limited to, thermal stress, vibrational stress, shock stress, gravitational G-force loading, moisture and/or contaminant exposure, as well as changes in the material due to age and use, such as the growth of tin whiskers, metal migration and delamination of materials. These faults can occur individually and/or in rapid succession on any chassis or backplane circuit or weapon system EWIS. Fault durations range in time from nanoseconds to milliseconds and have variable impedances. These circuit path disruptions are frequently caused by cracked solder joints, intermittent coax lines (e.g., shield corrosion, damaged center conductor, etc.), broken, cracked, or frayed wires, loose clamps, and unsoldered pins. These circuit path disruptions often cause functional failures/faults in LRU/WRA chassis and backplanes or weapon system EWIS whose root cause(s) cannot be detected and isolated using traditional ATE and troubleshooting processes. Lacking the ability to detect and isolate intermittent failures and provide environmental stimuli during the test and repair process, such assets are commonly reported as no-fault-found (NFF) or as one of the quasi-NFF repair codes (e.g., cannot duplicate, re-test OK, beyond the capability of maintenance, disassemble-clean-reassemble, etc.).

6.5.11 Intermittent fault susceptibility. Depending on the detection and isolation classification of the category (see 1.2) of the intermittent diagnostic equipment, the LRU/WRA or weapon system EWIS intermittent fault susceptibility should be considered in regard to appropriate maintenance action. The diagnostic equipment may detect intermittent faults within the LRU/WRA or weapon system EWIS that may not affect its operation. For example, in the weapon system EWIS carrying electrical power from the power source to the utilization equipment, the utilization equipment is designed to operate with power interrupts up to 50 milliseconds (see

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MIL-STD-704) without malfunction. Although the intermittent fault may not result in utilization equipment malfunction, the intermittent fault is often an early warning of a deteriorating condition of the weapon system EWIS, such as conductor strand breakage, chafing, corrosion, etc. All conductive paths in LRU/WRA chassis/backplanes and weapon system EWIS are susceptible to intermittent faults. Intermittent faults may not affect the functionality of the LRU/WRA or weapon system EWIS but may be an indicator of future impending failures related to corrosion, frayed insulation, broken wire strands, cold/cracked solder joints, etc.

6.5.12 Line Replaceable Unit (LRU). LRU is a modular component of an airplane, which is designed to be replaced quickly at an operating location. An LRU is usually a sealed unit such as a radio or other auxiliary equipment. LRUs are typically assigned logistics control numbers (LCNs) or work unit codes (WUCs) to manage logistics operations.

6.5.13 Long duration fault. A long duration fault is a fault which has a duration between 501 microseconds and 5 milliseconds and typically has a resistance of 501 ohms or more.

6.5.14 Open. An open, commonly referred to as an open circuit, is an abnormal or unintended loss of continuity that can occur in one or more conductive paths in the LRU/WRA chassis/backplane or weapon system EWIS. This results in an interruption or the loss of electric current in the case of a power circuit or an abnormal interruption or loss of signal in one or more conductive paths. The impedance of this conductive path is equal to the total impedance of the abnormal/unintended conductive path. The open impedance of the abnormal/unintended conductive path is defined as greater than 10 kilohms.

6.5.15 Shop Replaceable Unit /Shop Replaceable Assembly (SRU/SRA). The SRU is a modular component of an aircraft, ship, or spacecraft that is designed to be replaced by a technician at a backshop. Repair at backshops is known as intermediate-level (I-level) maintenance. SRUs are similar in nature to LRUs, but rather than being complete functional units, they represent component functions, such as circuit card assemblies, of a larger LRU. SRUs are typically assigned LCNs or WUCs to manage logistics operations. Shop Replaceable Assembly is a generic term that includes all replaceable packages within a WRA, including the chassis and wiring as a unit. See MIL-HDBK-2165.

6.5.16 Short. A short circuit, commonly referred to as a short, is an abnormal or unintended connection between two conductive paths in the LRU/WRA chassis/backplane or weapon system EWIS. Its existence can/will disrupt the operation of any of the contributing shorting paths. A short circuit condition may result in an increased electric current/over current in the case of a power circuit or an abnormal or unintended signal on two or more conductive paths. A short can be considered as an additional electrical path between adjacent electrical path(s) that is not designed to be part of, and its existence can/will disrupt the operation of any of the contributing shorting paths. For the purposes of this specification, the 'hard short' impedance of an abnormal/unintended conductive path is defined as less than 10 ohms; however, abnormal conductive paths with higher impedance values (i.e., hundreds or thousands of ohms) can cause equipment malfunctions and/or be an indication of degraded EWIS.

6.5.17 Short duration fault. A short duration fault is a fault with a duration under 100 nanoseconds and typically has a resistance under 10 ohms.

6.5.18 Test Program Set (TPS). A generic term that refers to the combination of interface cables/hardware and UUT-specific software/mapping that is required to test any LRU/WRA or EWIS subsystem.

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6.5.19 Weapon Replaceable Assembly (WRA). A generic term that includes all replaceable packages of a system installed in that system with the exception of cables, mounting provisions, and fuse boxes or circuit breakers. See MIL-HDBK-2165.

6.5.20 Weapon system. A combination of one or more weapons with all related equipment, materials, services, personnel, and means of delivery required for self-sufficiency. Weapon systems include the weapon system or military equipment, ground vehicles (wheeled, tracked, etc.), missiles, ships, air, space vehicles, etc. Intermittent faults may occur in any electrical components with electrically conductive paths, such as LRU/WRAs (military applications) or commercial electrical components. These electrical components, when installed in weapon systems or military equipment, may experience temperature, vibration, and shock environmental conditions that affect their functional operation.

6.6 Standard interface connector. The standard TPS mating interface test adapter cables use MIL-DTL-38999 series III connectors which are scoop-proof and environmental type. The plug part number is D38999/26WJ35PN using a typical back shell, with part number is M85049/18-25W06. The receptacle/mating side of the TPS adapter cable interface uses a D38999/20WJ35SN receptacle. The 128 contacts are standard SAE AS39029 configuration, size 22D with sockets on the receptacle side and pins on the plug side. The keying of the connector is Normal (none/default). Figure 2 shows the insert configuration for the 128-contact MIL-DTL-38999 connector used for the receptacle side.

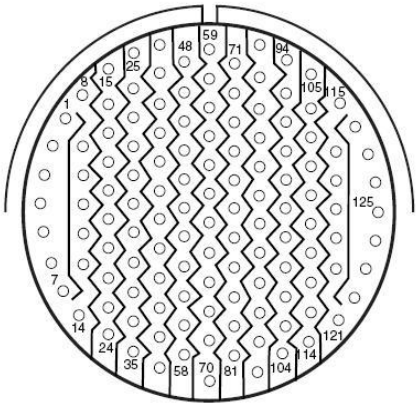


FIGURE 2. Standard interface test adapter connector.

6.7 Government validation. The diagnostic equipment may be tested in a laboratory environment by the Government using an Intermittent Fault Emulator (IFE). The emulator is capable of generating multiple intermittent faults to simulate 256 LRU/WRA conductive paths. The emulator is capable of generating controlled intermittent faults of various durations from 100 nanoseconds to one second at pseudo-random time intervals. See MIL-HDBK-527 for guidance on the use and application of the IFE, unless otherwise specified in the applicable specification sheet.

6.8 Nature of intermittent fault. Troubleshooting strategies for intermittent faults within EWIS often vary based on whether the intermittent fault is an intermittent open or an intermittent short; in general, troubleshooting for intermittent opens is usually localized to connectors/terminals/contacts/etc., while intermittent shorts may occur anywhere along the entire length of a wire run. Additionally, shorts to the ground plane of the UUT often occur due to conditions such as chafing on conductive structures, including harnesses chafing against fixed

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points or internal chafing within shielded harnesses. As such, differentiation between intermittent opens and intermittent shorts as a function of the diagnostic equipment is crucial, as it will inform the user with respect to where the defect root cause identification should begin. Given the challenges in detecting short-duration intermittent faults, differentiation between intermittent opens and shorts does not necessarily have to be a primary function of the diagnostic equipment as long as a follow-on troubleshooting procedure is designed into the equipment. Such follow-on troubleshooting procedures can be used to provide supplemental information to the user in isolating short-duration intermittent faults, allowing for proper troubleshooting and corrective action. For example, since open circuits generally have rising impedance and short circuits generally have falling impedance, providing this type of information to the user will assist in root cause identification (see 3.10.1.1).

6.9 Physical characteristics. Diagnostic equipment design should consider the human factors engineering design principles of MIL-STD-1472 to ensure the equipment can be operated effectively and safely in its intended environment by appropriately trained personnel. All components should be selected and located for maximum ease of operation, inspection, and maintenance.

6.10 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.11 Subject term (key word) listing.

Connectors
EWIS
Intermittent
Long Duration
Shorts
Terminal Boards
Wire Harness

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APPENDIX A

VIBRATION STIMULATION

A.1 SCOPE

A.1.1 Scope. This appendix details ways to isolate failures related to vibration. This appendix is not a mandatory part of the specification. The information contained herein is for guidance only.

A.2 DETERMINING CAUSES OF INTERMITTENT FAILURES

A.2.1 Introduction. Each LRU/WRA is different in its function and operational environment. As a result, no single test method or procedure can adequately replicate an intermittent fault occurrence for all LRUs/WRAs. A careful review of the nature of the failure and the operational conditions under which the failure occurred is required. The following steps are recommended when by careful analysis, it is determined that the failures occur during ground or flight operating conditions, and the operating temperature does not appear to be contributing to the occurrence of the failures.

A.2.2 Typical resulting effects. The following is a list of typical resulting effects of vibration-induced problems (this list is not all-inclusive):

a. Chafed wiring.

b. Loose fasteners/components.

c. Intermittent electrical contacts.

d. Electrical shorts.

e. Deformed seals.

f. Failed components.

g. Optical or mechanical misalignment.

h. Cracked and/or broken structures.

i. Migration of particles and failed components.

j. Particles and failed components lodged in circuitry or mechanisms.

k. Excessive electrical noise.

l. Fretting corrosion in bearings.

A.2.3 Operational vibration environment. A review should be conducted of technical manuals, operating manuals, and any available information which provides insight into the operational vibration of the LRU/WRA. As much as practical, this information should be used to tailor a vibration envelope for vibrating the LRU/WRA while troubleshooting the LRU/WRA for intermittent faults. It is not necessary to vibrate the LRU/WRA at full qualification levels, which may induce additional failure modes. It is recommended that where the operational vibration environment is not known, the qualification vibration test levels may be reduced by a factor of eight and used during troubleshooting of the LRU/WRA. The intent is to subject the LRU/WRA to a vibration level high enough to stimulate the intermittent fault but not reduce the operational life of the LRU/WRA.

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A.2.3.1 Unknown operational limits. Where the vibration operational limits are unknown, MIL-STD-810, Test Method 514 should be used to tailor the LRU/WRA vibration stimulation test sequence as a function of the life cycle environments of the LRU/WRA:

A.2.3.1.a General. The accumulated effects of vibration-induced stress may affect LRU/WRA performance under other environmental conditions such as temperature, altitude, humidity, leakage, or electromagnetic interference/electromagnetic compatibility. When evaluating the cumulative environmental effects of vibration and other environments, expose the LRU/WRA to all environmental conditions, with vibration testing generally performed first. If another environment (e.g., temperature cycling) is projected to produce damage that would make the LRU/WRA more susceptible to vibration, perform tests for that environment before vibration tests. For example, thermal cycles might initiate a fatigue crack that would grow under vibration or vice versa.

A.2.3.1.b Unique to this method. Generally, expose the LRU/WRA to the sequence of individual vibration tests that follow the sequence of the life cycle. For most tests, this can be varied if necessary to accommodate test facility schedules or for other practical reasons. Complete any maintenance associated preconditioning prior to tests representing mission environments. Perform tests representing critical end-of-mission environments last.

A.2.4 Operational level vibration. Functional testing is conducted to verify that the LRU/WRA functions as required while exposed to an operational level of vibration. This level of vibration exposure is considerably lower than the qualification level exposure. The intent is to subject the LRU/WRA to a vibration level high enough to stimulate the intermittent fault(s) but not reduce the operational life of the LRU/WRA. Functional level vibration testing in accordance with MIL-STD-810, Test Method 514 is recommended. Typically, "operating" and "non-operating" envelopes are established. Tailor functional tests to accommodate non-operating portions by modifying required functional monitoring requirements as appropriate (see A.2.3).

A.2.5 Operational Tailoring Guidance for vibration exposure. MIL-STD-810, Test Method 514 Annex D provides information intended to be useful in determining the vibration levels and durations of operational environmental life cycle events and in defining the tests necessary to develop materiel to operate in and survive these environments.

A.2.5.1 Category 12 - Fixed-wing aircraft-jet aircraft. The vibration environment for materiel installed in jet aircraft (except engine-mounted) and gunfire-induced stems from four principal mechanisms. These are (1) engine noise impinging on aircraft structures; (2) turbulent aerodynamic flow over external aircraft structures, (3) turbulent aerodynamic flow and acoustic resonance phenomena within cavities open to the external airflow, particularly open weapon bays, and (4) airframe structural motions due to maneuvers, aerodynamic buffet, landing, taxi, etc. Vibration can also be produced by installed materiel items. These vibrations are generally important only locally at or near the source and may not be significant even in that local area.

A.2.5.2 Category 13 - Propeller aircraft. The vibration environment for materiel installed in propeller aircraft (except engine-mounted and gunfire induced) is primarily propeller induced. The vibration frequency spectra consist of a broadband background with superimposed spikes. The spikes are more nearly sinusoidal than narrowband and are best represented utilizing sine-on-random vibration. The background spectrum results from various random sources combined with many lower-level periodic components due to the rotating elements (engines, gearboxes, shafts,

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etc.) associated with turboprops. The spikes are produced by the passage of pressure fields rotating with the propeller blades. These occur in relatively narrow bands centered on the propeller passage frequency (number of blades multiplied by the propeller rpm) and harmonics.

A.2.5.3 Category 14 - Rotary wing aircraft - helicopter. Helicopter vibration (for engine-mounted materiel below and for gunfire-induced vibration) is characterized by dominant peaks superimposed on a broadband background. The peaks are sinusoids produced by the major rotating components (main rotor, tail rotor, engine, gearboxes, shafting, etc.). The peaks occur at the rotation speed (frequency) of each component (i.e., 1P for the main rotor, 1T for the tail rotor, and 1S where S designates a locally predominate rotating element) and harmonics of these speeds (e.g., 2P, 3P, 4P). The broadband background is a mixture of lower amplitude sinusoids and random vibrations due to sources such as aerodynamic flow noise. Vibration levels and spectrum shapes vary widely between helicopter types and throughout each helicopter, depending on the strength and location of sources and the geometry and stiffness of the structure. Thus, the need for measured data is acute.

A.2.5.4 Category 15 - Aircraft stores - assembled, jet aircraft. Assembled jet aircraft stores may encounter three distinct vibration environments: external captive carriage, internal captive carriage, and free flight.

A.2.5.5 Category 16 - Aircraft stores - materiel, jet aircraft. Materiel installed within a jet aircraft store will experience the store vibration. The input exposure levels for materiel within the store are essentially the same as the response levels of the store. If gunfire, cavity resonance, buffet-maneuver, and free-flight conditions occur for the store, the materiel will also be exposed to these conditions.

A.2.5.6 Category 17 - Aircraft stores - assembled/materiel, propeller aircraft. There is no known source of general guidance or measured data for the vibration of propeller aircraft stores (except gunfire induced). However, since the excitation sources are the same, it seems likely that store vibration will be similar to that of the carrying aircraft. Maneuver buffet vibration experienced by stores of highly maneuverable propeller aircraft should be similar to that experienced by jet aircraft stores.

A.2.5.7 Category 18 - Aircraft stores - assembled/materiel, helicopter. Complex periodic waveforms characterize the service environment encountered by assembled stores externally carried on helicopters. Unlike stores carried on fixed-wing aircraft, externally mounted helicopter stores receive little aerodynamic excitation, particularly when compared with the rotor-induced vibration. Thus, most of the vibratory energy reaches the store and materiel through the attachment points between the aircraft and the store. Some excitation, however, is added along with the entire store structure due to periodic rotor-induced pressure fluctuations. The result is a complex response unique to the particular aircraft-store configuration. Therefore, a realistic definition of the environment depends almost totally upon the use of in-flight vibration measurements. For stores exposed to gunfire, refer to MIL-STD-810, Test Method 519.

A.2.5.8 Category 19 - Missiles - Tactical missiles (free flight). There is no known source of general guidance or measured data for tactical missile carriage or launch vibration environments.

A.2.5.9 Category 20 - Ground vehicles - ground mobile. The ground mobile environment consists of broadband random vibration with peaks and notches. These peaks and notches are

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considerably higher and lower than the mean level. Terrain, road, and surface discontinuities, vehicle speed, loading, structural characteristics, and suspension system all affect this vibration. Gunfire criteria are not applicable since it is based on the response of aircraft-type structures that are significantly different than ground vehicle structures.

A.2.5.10 Category 21 - Watercraft - marine vehicles. Marine vibration spectra have a random component induced by the variability of cruising speeds, sea states, maneuvers, etc., and a periodic component imposed by propeller shaft rotation and hull resonance. Materiel mounted on masts (such as antennas) can be expected to receive higher input than materiel mounted on the hull or deck. The overall ship's structure, materiel mounting structure, and materiel transmissibility (amplifications) greatly affect materiel vibration. Development of shipboard materiel should address both the levels of environmental inputs and the coincidence of materiel/mounting resonances and input frequencies.

Note: For US Navy applications, refer to MIL-STD-810 Method 528

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TEMPERATURE STIMULATION

B.1 SCOPE

B.1.1 Scope. This appendix deals with ways to isolate faults related to operating temperature. This appendix is not a mandatory part of the specification. The information contained herein is for guidance only.

B.2 DETERMINING CAUSES OF INTERMITTENT FAILURES

B.2.1 Introduction. Each LRU/WRA is different in its function and operational environment. As a result, no single test method or procedure can adequately replicate an intermittent fault occurrence for all LRUs/WRAs. A careful review of the nature of the failure and the operational conditions under which the failure occurred is required. The following steps are recommended when by careful analysis, it is determined that the failures occur during ground or flight operating conditions, and the operating temperature appears to be contributing to the occurrence of the failures.

B.2.2 Typical resulting effects. The following is a list of typical resulting effects of temperature-induced problems (this list is not all-inclusive):

a. Binding or slackening of moving parts.

b. Deformation or fracture of components.

c. Cracking of surface coatings.

d. Leaking of sealed compartments.

e. Failure of insulation protection.

f. Differential contraction or expansion rates or induced strain rates of dissimilar materials.

g. Intermittent electrical contacts.

h. Electrical shorts/opens.

i. Failed components.

j. Changes in electrical and electronic components.

k. Electronic or mechanical failures due to rapid water or frost formation.

l. Excessive static electricity.

B.2.3 Operational temperature environment. A review should be conducted of technical manuals, operating manuals, and any available information that provides insight into the operational temperature environment of the LRU/WRA. As much as practical, this information should be used to tailor a temperature cycling profile for temperature stressing the LRU/WRA while troubleshooting the LRU/WRA for intermittent faults. It is not necessary to temperature cycle the LRU/WRA at full qualification levels, which may induce additional failure modes. It is recommended that where the operational temperature test levels are not known that the qualification temperature levels during troubleshooting of the LRU/WRA be reduced in order to not overstress the LRU/WRA. The intent is to subject the LRU/WRA to a temperature level low/high enough to stimulate the intermittent fault but not reduce the operational life of the LRU/WRA.

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B.2.3.1 Unknown temperature operational limits. Where the temperature operational limits are unknown, MIL-STD-810, Test Method 503, should be used to tailor the LRU/WRA temperature stimulation test sequence as a function of the life cycle environments of the LRU/WRA. It should be noted that Test Method 503 is a temperature shock test method and should be tailored to represent the operational temperature changes that the LRU/WRA is exposed to in its operating environment. Procedure I-C, multi-cycle shocks from constant extreme temperature, or Procedure I-D, shocks to or from controlled ambient temperature, is recommended depending on the operational environment of the LRU/WRA.

B.2.3.1.1 Information collection. During the temperature cycling testing, the following information should be collected:

a. Record of chamber temperature versus time conditions.

b. Test item temperatures (measured locations).

c. Transfer times (e.g., "door open" to "door closed").

d. Duration of each exposure.

e. Conductive path intermittent fault location.

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TEMPERATURE/VIBRATION STIMULATION

C.1 SCOPE

C.1.1 Scope. This appendix deals with ways to isolate faults related to a combination of operating temperature and vibration. This appendix is not a mandatory part of the specification. The information contained herein is for guidance only.

C.2 DETERMINING CAUSES OF INTERMITTENT FAILURES

C.2.1 Introduction. LRU/WRAs are different in their function and operational environments. As a result, no single test method or procedure can adequately replicate an intermittent fault occurrence for all LRUs/WRAs. A careful review of the nature of the failure and the operational conditions under which the failure occurred is required. The following steps are recommended when by careful analysis, it is determined that the failures occur during ground or flight operating conditions.

C.2.2 Typical failures. Temperature, humidity, vibration, and altitude can combine synergistically to produce the following failures. Although altitude is included in the following discussion typically in regard to LRU/WRA operating environment, it mainly impacts cooling and is a function of temperature. Typically combined environmental test facilities do not include altitude test capability. The following examples of failures due to temperature, humidity, or vibration are not intended to be comprehensive:

a. Shattering of optical material (Temperature/Vibration/Altitude).

b. Binding or loosening of moving parts (Temperature/Vibration).

c. Separation of constituents (Temperature/Humidity/Vibration/Altitude).

d. Performance degradation in electronic components due to parameter shifts (Temperature/Humidity).

e. Electronic optical (fogging) or mechanical failures due to rapid water or frost formation (Temperature/Humidity).

f. Differential contraction or expansion of dissimilar materials (Temperature/Altitude).

g. Deformation or fracture of components (Temperature/Vibration/Altitude).

h. Cracking of surface coatings (Temperature/Humidity/Vibration/Altitude).

i. Leakage of sealed compartments (Temperature/Vibration/Altitude).

j. Failure due to inadequate heat dissipation (Temperature/Vibration/Altitude).

C.2.3 Combined forcing functions. A review should be conducted of technical manuals, operating manuals, and any available information. Apply the tailoring process in MIL-STD-810 to determine where these combined forcing functions of temperature, humidity, vibration, and altitude are foreseen in the LRU/WRA operational environment. Use this method only if the proper engineering has been performed, such that the environmental stresses associated with the individual methods are encompassed by the combined test. If appropriate, tailor storage thermal environments into the combined environmental cycle; or perform them as separate tests, using the individual test methods. Use the following to aid in selecting this

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method and placing it in sequence with other methods.

C.2.3.1 Unknown operational limits. Where the temperature/vibration/humidity/altitude operational limits are unknown, MIL-STD-810, Test Method 520 should be used to tailor the LRU/WRA temperature/vibration stimulation test sequence as a function of the life cycle environments of the LRU/WRA:

C.2.3.1.1 Vibration. Four vibration profiles may be used:

a. A random test profile with the following parameters is an example of temperature/vibration:

b. A 20-800 Hz random profile where $G^2/Hz = 0.0051282$ and G RMS = 2.

c. A 20-300 Hz sine sweep that runs for 3:54 where G RMS = 2.

d. A 20-300 Hz sine sweep that runs for 0:15 where G RMS = 2.

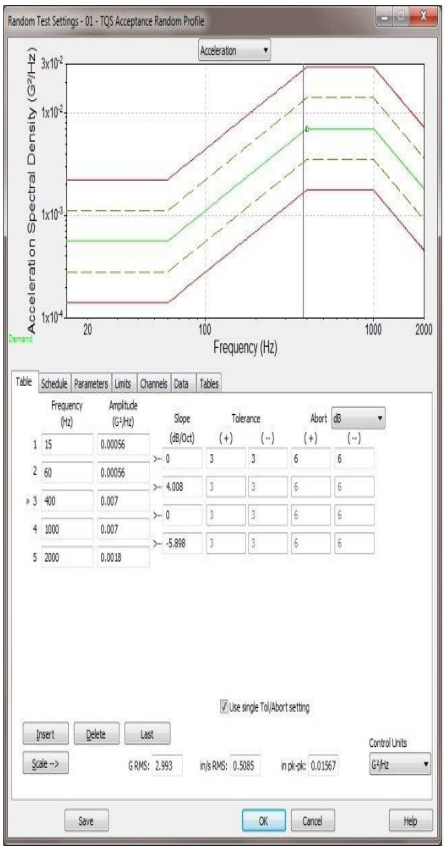


FIGURE C-1. Example random vibration test profile.

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C.2.3.1.2 Temperature.

a. Initially, the vibration profiles are run at room temperature, approximately 75 °F (24 °C).

b. The temperature is then dropped to 32 °F (0 °C) at 41 °F (5 °C)/minute, and vibration profiles are run as the temperature drops.

c. The temperature is then held at 32 °F (0 °C) for a 15-minute soak time, during which time the vibration profiles are run.

d. The temperature is then dropped to -40 °F (-40 °C) at 41 °F (5 °C)/minute, and vibration profiles are run as the temperature drops.

e. The temperature is then held at -40 °F (-40 °C) for a 15-minute soak time, during which time the vibration profiles are run.

f. The temperature is then raised to 158 °F (70 °C) at 41 °F (5 °C)/minute, and vibration profiles are run as the temperature rises.

g. The temperature is then held at 158 °F (70 °C) for a 15-minute soak time, during which time the vibration profiles are run.

h. Finally, the temperature is returned to room temperature at 41 °F (5 °C)/minute, and vibration profiles are run as the temperature falls.

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CONCLUDING MATERIAL

Custodians:
Army - MI
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS
(Project 6625-2021-006)

Review activity:
Air Force - 170

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